

Juan Pablo Valenzuela Castro

AI Software Engineer · Full Stack Developer · Cloud ML Systems

valenzuelacastrojuanpablo@gmail.com · [linkedin.com/in/iTzTorero](https://www.linkedin.com/in/iTzTorero) · itztorero.dev · github.com/iTzTorero ·
Guasave, Sinaloa. México

PROFESSIONAL SUMMARY

Full Stack Software Engineer with 3+ years of experience building end-to-end SaaS products and scalable cloud-native systems on Google Cloud Platform, with artificial intelligence integrated as a functional layer, not an add-on. I build both the frontend (Angular, JavaScript/TypeScript) and the backend (Node.js, Python, event-driven microservices), and integrate LLMs through function calling, RAG, and workflow orchestration with deterministic guardrails. Proven track record taking complete products to production: architecture, APIs, recurring billing with Stripe, authentication, and cloud operations.

TECHNICAL SKILLS

AI & LLMs: OpenAI API · LLM Integration · Function Calling / Tool Use · RAG Architectures · Multi-Agent Systems · Prompt Engineering · AI Workflow Orchestration

Backend & APIs: Python · FastAPI · Node.js · REST APIs · Microservices · Async/Event-Driven Architecture · Docker · CI/CD

Cloud (GCP): Cloud Run · Pub/Sub · BigQuery · Firestore · Cloud Storage · Firebase · GCP IAM · Cloud Functions

Frontend: JavaScript · TypeScript · Angular · Vue 3 · Nuxt · Ionic · SPA · SCSS · PWA · Capacitor

Machine Learning: scikit-learn · MLP · SVM · Computer Vision · Multispectral Imaging · ML Pipelines · Model Deployment

Data & DevOps: BigQuery · Firestore · Redis · ETL Pipelines · Linux · Git · Docker · Monitoring

EXPERIENCE

Software Engineer · Monteli LLC

Jan 2023 – Present

- Designed and deployed scalable data ingestion and transformation pipelines on GCP (Cloud Run, Pub/Sub, Python, Node.js) serving multiple SaaS products and enterprise clients.
- Designed and deployed event-driven microservices supporting LLM-powered workflows and large-scale automated data processing.
- Built and maintained production REST APIs for AI/ML inference services, enabling real-time processing and integration with external systems.
- Integrated the OpenAI API into production workflows: structured outputs, tool use, and orchestration of multi-step automation pipelines.
- Implemented RAG-ready architectures for context-aware responses grounded in structured and unstructured data sources.
- Owned the full backend lifecycle: API design, Dockerization, CI/CD pipelines, monitoring, and production support.
- Designed secure integrations with external services: OAuth flows, JWT authentication, and third-party API communication.

FEATURED PROJECTS

AgendaBot Pro — WhatsApp Appointment Scheduling SaaS | *Angular · TypeScript · Node.js · Firestore · Cloud Functions · GPT (Function Calling) · WhatsApp Cloud API · Stripe · CFDI · GCP*

- Multi-tenant SaaS for the Mexican/LATAM market that lets patients book, reschedule, and cancel appointments entirely from WhatsApp, with no extra app.
- AI agent powered by GPT function calling with deterministic guardrails: the model only invokes backend tools and never writes directly to the database.
- Built the product end to end: frontend, backend, recurring billing with Stripe, and Mexican e-invoicing (CFDI).

Balltics — Baseball League Management Platform | *Angular 17 · TypeScript · Angular Signals · RxJS · Firebase · Firestore · Cloud Functions · IndexedDB · Stripe · GCP*

- Multi-tenant SaaS to run amateur leagues end to end: a real-time public site (standings, leaders, full box scores) plus an admin suite.
- Live pitch-by-pitch scoring with offline-first support (IndexedDB) and recurring billing with Stripe.

FinOps Multi-Agent Assistant | *Angular · TypeScript · GCP · Firebase · LLM · RAG · Multi-Agent*

- Designed a multi-agent architecture with task routing, memory handling, and tool execution for finance workflow automation.
- Implemented a RAG pipeline for context-aware responses grounded in structured financial datasets.

AgriLumens — Precision Agriculture ML Platform | *FastAPI · Python · GCP · OpenDroneMap · Computer Vision*

- Built a modular ML inference pipeline over multispectral drone imagery: preprocessing, feature extraction, and prediction of inorganic nitrogen and organic matter in soils.
- Technical foundation of master's thesis at IPN: MLP and SVM models applied to spectral imagery with statistical validation (Walkley & Black method).

Big Data Platform (ETL + SaaS Products) | *Node.js · Python · GCP · Cloud Run · Pub/Sub · Firestore*

- Core data platform that ingests, normalizes, and serves datasets consumed by multiple SaaS products and enterprise clients.

Nauti Test Pro — Mobile Exam Prep Platform | *Vue 3 · Ionic · Firebase · Capacitor · Cloud Functions*

- Mobile-first PWA for nautical license exam prep with timed tests, progress tracking, and subscription-based access.

EDUCATION

Master of Science in Natural Resources and Environment · IPN — National Polytechnic Institute

Jan 2024 – Jan 2026

Research: Estimation of inorganic nitrogen and organic matter in agricultural soils using multispectral imagery and artificial intelligence models

Software Engineering · ITSON — Sonora Institute of Technology

Aug 2019 – May 2023